



Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year: 2025), B.Sc. in CSE (Day)

Lab Report: 03

Course Title: Data Mining Lab
Course Code: CSE 436
Section: 212_D2

Lab Manual 04

Statistical Analysis and Data Visualization

Student Details

Name	ID
Md. Almahmud	212002142

Submission Date : 30-04-2025
Course Teacher's Name : Md. Atikuzzaman

Lab Report Status

Marks :Signature.....

Comments :Date :.....

1. TITLE OF THE LAB EXPERIMENT:

Statistical analysis and data visualization using Python. Use “Titanic Dataset” or “titanic = sns.load_dataset('titanic')”.

2. OBJECTIVES/AIM:

- To perform basic statistical analysis and visualize patterns in the Titanic dataset using Python libraries such as Pandas, Seaborn, and Matplotlib.

3. MOTIVATION:

- Statistical analysis helps us understand the data's central tendencies, spread, and relationships between variables.
- Data visualization reveals hidden patterns and trends that are not easily visible in raw data.

4. PROCEDURE:

- Load the Titanic dataset using Seaborn's built-in function.
- Perform exploratory data analysis (EDA) to understand the data structure.
- Generate descriptive statistics (mean, median, mode, standard deviation, etc.).
- Use plots (bar, histogram, boxplot, heatmap, etc.) to visualize distributions and relationships.
- Interpret findings from the analysis and visuals.

5. CODE:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

titanic = sns.load_dataset('titanic')
print(titanic.head())
print(titanic.info())
print(titanic.describe(include=['object', 'category']))

print("\n Countplot - Survivors vs Non-survivors: ")
plt.figure(figsize=(6, 4))
sns.countplot(x='survived', data=titanic)
plt.title('Survival Count')
plt.xticks([0, 1], ['Not Survived', 'Survived'])
plt.show()

print("\nHow Many Survival by Sex: ")
plt.figure(figsize=(6, 4))
sns.countplot(x='sex', hue='survived', data=titanic)
plt.title('Survival by Gender')
plt.legend(['Not Survived', 'Survived'])
```

```
plt.show()

print("\nSurvival by Age: ")
plt.figure(figsize=(8, 6))
sns.histplot(data=titanic, x='age', hue='survived', kde=True)
plt.title('Survival by Age')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()

print("\nBoxplot: ")
plt.figure(figsize=(6, 4))
sns.boxplot(x='class', y='fare', data=titanic)
plt.title('Fare Distribution by Passenger Class')
plt.show()

print("\nHeatmap - Correlation Matrix: ")
plt.figure(figsize=(10, 6))
sns.heatmap(titanic.corr(numeric_only=True), annot=True, cmap='coolwarm',
fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()
```

6. OUTPUT:

Dataset:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

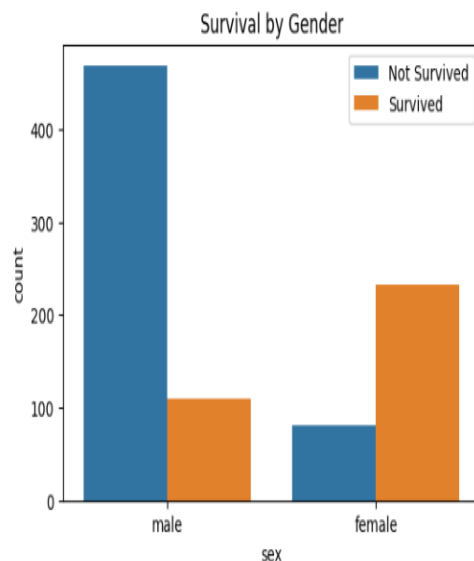
	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Dataset Information:

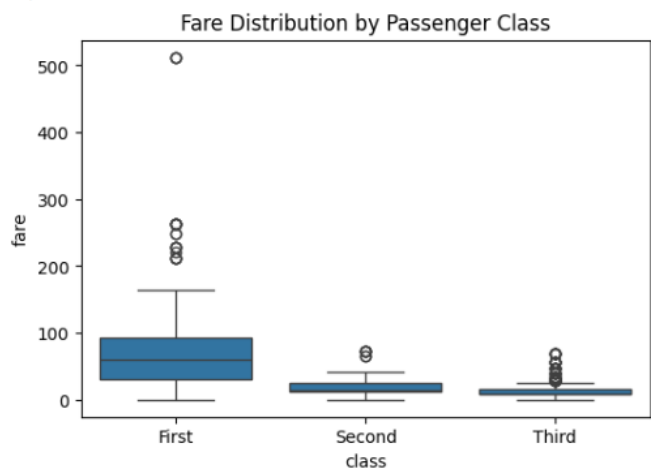
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male  891 non-null    bool
11  deck        203 non-null    category
12  embark_town 889 non-null    object
13  alive       891 non-null    object
14  alone       891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

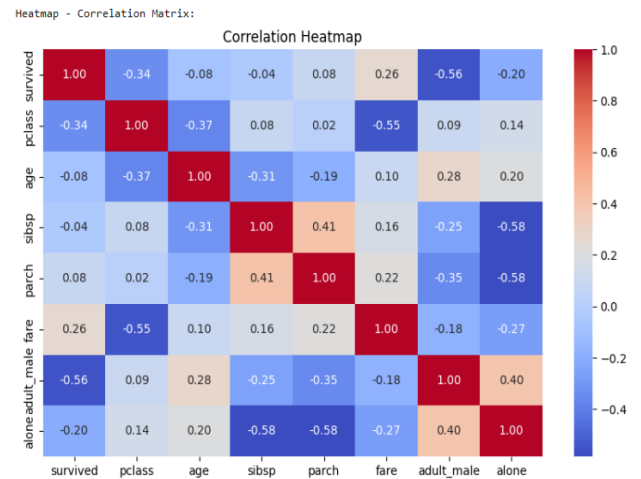
Dataset Description:

	sex	embarked	class	who	deck	embark_town	alive
count	891	889	891	891	203	889	891
unique	2	3	3	3	7	3	2
top	male	S	Third	man	C	Southampton	no
freq	577	644	491	537	59	644	549



Boxplot:





7. DISCUSSION:

The Titanic dataset includes important passenger information such as age, gender, class, fare, and whether they survived or not. To understand the data better, descriptive statistics were used, which gave insights like the average age of passengers, how fares were spread out, and how many people traveled in each class. Cleaning steps such as handling missing values (using forward fill), removing duplicates, and standardizing column names were also done to prepare the dataset for analysis and modeling.

Visualization played a key role in exploring the dataset. Heatmaps were used to check for missing values and correlations between variables. Boxplots helped detect outliers in numerical features like age and fare. Histplots showed the distribution of age and fare, while countplots revealed class-wise or gender-wise passenger counts. Most importantly, visual tools were used to understand survival patterns—like how survival rates differed by gender, age group, or passenger class—giving us a clearer picture of the data.